25900149 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCPEG100 - DRILLING, TRANSPORTATION ANDREFINING TECHNIQUES

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple choice questions

Answer all questions

Each question carries **1** marks

1. Select which of the following technologies helps to detect leaks and monitor the condition of offshore rigs in real-time. [K] / [CO1]

   a). Drones and remote sensing systems

   b). Manual inspection by workers

   c). Automated drilling rigs

   d). Seismic wave monitoring

2. Identify the purpose of casing in an oil well. [U] / [CO1]

   a). To drill faster

   b). To prevent the collapse of the wellbore

   c). To remove drilling fluid

   d). To increase production rate

3. What environmental risk is associated with improper disposal of oil field water? [K] / [CO1]

   a). Increase in crude oil prices

   b). Increase in drilling efficiency

c). Contamination of groundwater d). Reduction in oil density

4. Connate water brines are classified as : [U] / [CO1]

a). Meteoric brines

b). Fossil seawater trapped in sediments

c). Hydrothermal brines

d). Volcanic brines

5. Which of the following best defines a ‘complex refinery’? [U] / [CO3]

a). A refinery that distills crude oil without further processing

b). A refinery that includes catalytic cracking and reforming units

c). A refinery that produces only lubricating oil

d). A refinery that specializes in bitumen emulsification

6. Relate from the following the importance of sweetening process in refinery operations. [U] / [CO4]

a). To increase the boiling point of crude oil

b). To remove sulfur compounds and improve fuel quality

c). To separate heavier hydrocarbons

d). To enhance lubricating oil viscosity

7. A plant requires a heat exchanger for a high-pressure system with temperatures exceeding 260°C. Which exchanger is most suitable? [U] / [CO5]

a). Plate heat exchanger

b). Spiral heat exchanger

c). Shell and tube heat exchanger

d). Double-pipe heat exchanger

8. Identify the natural method of oil recovery. [U] / [CO1]

a). Gas flood method b). Water flood method

c). Dissolved gas drive d). Chemical injection

9. How would you use the viscosity test to assess the performance of lubricants and oils in practical applications? [K] / [CO6]

a). The temperature at which the fluid freezes

b). The resistance of the fluid to flow

c). The ability of the fluid to vaporize

d). The concentration of dissolved gases

10. The standard method used to determine the cloud point: [U] / [CO6]

a). ASTM D97 b). ASTM D86

c). ASTM D2500 d). ASTM D445

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **3** marks

11. Outline the difference between offshore and onshore drilling. [K] / [CO1]

12. Discuss the applications of nitrogen blanketing. [U] / [CO2]

13. How does the presence of sulfur analyze the quality and usability of petroleum products? [An] / [CO4]

14. Describe the differences between LAN (Low Aromatic Naphtha) and HAN (High Aromatic Naphtha) based on composition and applications. [U] / [CO4]

15. Describe how water circulation influence hydrocarbon migration. [U] / [CO1]

16. Illustrate the softening point test of bitumen. [An] / [CO6]

[3x4 = 12]

Part C

Short Essay Questions

Answer any **4** questions

Each question carries **7** marks

17. Illustrate the environmental concerns associated with extracting petroleum from alternative sources. [A] / [CO1]

18. Describe the advantages and disadvantages of transporting oil and gas by pipeline, and railway. [U] / [CO2]

19. Give an account on composition of different distillates -Kerosene, Diesel, Fuel oil, Bitumen. [U] / [CO4]

20. Compare and contrast Natural Rubber Modified Bitumen and Bitumen Emulsion in terms of performance, durability, and environmental impact. [An] / [CO4]

21. Discuss in detail about PMC method of analysis. [U] / [CO6]

22. Examine the significance of density measurement in industrial applications. [An] / [CO6]

[7x4 = 28]